

A Geometric Multigrid Preconditioner for Shifted Boundary Method

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1 Introduction

2 Method formulation

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We consider the Poisson problem as a model problem:

$$-\Delta u = f \quad \text{in } \Omega, \quad (1)$$

$$u = g \quad \text{on } \Gamma, \quad (2)$$

where $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$) is a domain with boundary $\Gamma = \partial\Omega$ as depicted in Figure 1, f is a given source term, and g is the prescribed Dirichlet boundary condition.

To solve this problem numerically, we first formulate it in a weak sense. We seek a solution u in an appropriate function space, $H^1(\Omega)$, which consists of functions that are square-integrable and whose first derivatives are also square-integrable. Multiplying the equation by a test function $v \in H^1(\Omega)$ and integrating over Ω , we obtain:

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx - \int_{\Gamma} (\nabla u \cdot \mathbf{n}) v \, ds = \int_{\Omega} f v \, dx.$$

We next introduce a triangulation $\mathcal{T}_h$ consisting of quadrilateral (2D) or hexahedral (3D) elements of size h , and define a finite element space $\mathbb{V}_h \subset H^1(\Omega)$ using Lagrange polynomial elements of degree p . In classical finite element methods, the mesh conforms to the boundary (unlike the background mesh approach illustrated in Figure 1), and test functions v typically vanish on Γ to strongly enforce the Dirichlet condition $u = g$. When function spaces do not necessarily satisfy essential boundary conditions strongly, boundary conditions must be enforced weakly through additional integral terms on Γ .

Nitsche's method provides a way to weakly impose Dirichlet boundary conditions within a variational formulation without requiring the function space to satisfy the boundary conditions. It modifies the bilinear form by adding terms on the boundary Γ . The standard Nitsche formulation for the Poisson problem with Dirichlet boundary conditions $u = g$ on Γ seeks $u_h \in \mathbb{V}_h$ such that for all $v_h \in \mathbb{V}_h$:

$$\begin{aligned} \int_{\Omega} \nabla u_h \cdot \nabla v_h \, dx - \int_{\Gamma} (\nabla u_h \cdot \mathbf{n}) v_h \, ds - \alpha \int_{\Gamma} (\nabla v_h \cdot \mathbf{n}) u_h \, ds + \int_{\Gamma} \sigma u_h v_h \, ds = \\ = \int_{\Omega} f v_h \, dx - \int_{\Gamma} (\nabla v_h \cdot \mathbf{n}) g \, ds + \int_{\Gamma} \sigma_{\Gamma} g v_h \, ds. \end{aligned}$$

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Here, σ_Γ is a penalty parameter, typically chosen as $\sigma_\Gamma = \mathcal{O}(p^2 h^{-1})$ for mesh size h , and $\mathbf{n}$ is the outward unit normal vector to Γ . The choice of parameter $\alpha = 1$ leads to a symmetric formulation, while $\alpha = -1$ results in a non-symmetric formulation in which the penalty term can be skipped [2]

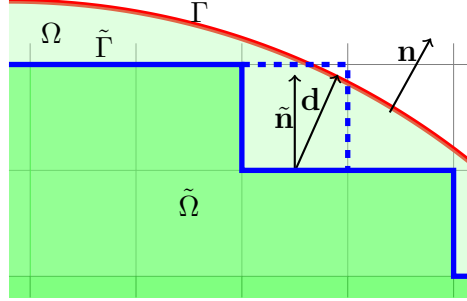


Figure 1: Schematic illustrating the background mesh, interior cells (green), the surrogate boundary $\tilde{\Gamma}$ (thick blue line) along the upper boundary of the interior cells, and the true boundary $\partial\Omega$ (red).

2.1 Shifted Boundary Method

In many applications, the domain Ω may have a complex geometry, making the generation of body-fitted meshes challenging. Non-body-fitted (unfitted) methods address this challenge by employing a background mesh $\mathcal{T}_h$ that does not conform to the boundary of Ω . The domain Ω is embedded within this background mesh, and cells of $\mathcal{T}_h$ are classified as active based on their intersection with Ω . The computational domain $\tilde{\Omega}$ is defined as the union of these active cells. In the original SBM formulation [5], only cells strictly contained in Ω were included, leading to the surrogate domain boundary depicted by the solid blue line in Figure 1. Consequently, the surrogate domain $\tilde{\Omega}$ is generally a subset of Ω , and its boundary $\tilde{\Gamma}$ does not coincide with the true boundary Γ .

Later extensions include intersected cells [8] with a volume fraction inside Ω greater than a threshold $\lambda \in [0, 1]$. In Figure 1, this corresponds to including additional cells as indicated by the dashed blue line. This approach reduces the distance between true and surrogate boundaries, but can lead to ill-conditioning.

To impose boundary conditions on $\tilde{\Gamma}$, the Dirichlet condition prescribed on the true boundary Γ is extrapolated to the surrogate boundary. This is typically accomplished by projecting points from $\tilde{\Gamma}$ onto the true boundary Γ along a suitable direction (e.g., the outward normal to $\tilde{\Gamma}$) and using a Taylor expansion to approximate the boundary values. This is accomplished via an *extension operator* $\mathcal{E}$, which maps functions defined on $\tilde{\Omega}$ to the surrogate boundary $\tilde{\Gamma}$.

We assume that the Dirichlet boundary condition g is given as a restriction of a function u^* defined on the entire domain Ω to the boundary Γ . While the choice of this function u^* (as an extension of g from Γ into Ω) is not unique, we take u^* to be equal to the solution u in the surrogate domain $\tilde{\Omega}$. For each point $\tilde{\mathbf{x}} \in \tilde{\Gamma}$, let $\mathbf{x} \in \Gamma$ be its closest point projection onto the true boundary, and let $\mathbf{d} = \mathbf{x} - \tilde{\mathbf{x}}$ be the shift vector. The function u is extended from Γ to $\tilde{\Gamma}$ using a Taylor expansion:

$$\mathcal{E}u^*(\tilde{\mathbf{x}}) = u^*(\mathbf{x}) - \mathbf{d} \cdot \nabla u(\tilde{\mathbf{x}}) + \dots$$

where $\mathcal{E}u^*$ denotes the extrapolated boundary condition. The function u^* is assumed to be smooth in a neighborhood of $\tilde{\Gamma}$, which allows for the Taylor expansion to be valid. By substituting it into the weak formulation, we obtain a variational formulation for the shifted boundary problem with the extrapolated boundary condition on $\tilde{\Gamma}$ enforced in a Nitsche-like manner; the

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weak formulation seeks $u_h \in V_h$ such that for all $v_h \in V_h$,

$$\begin{aligned} \int_{\tilde{\Omega}} \nabla u_h \cdot \nabla v_h \, dx - \int_{\tilde{\Gamma}} (\nabla u_h \cdot \tilde{\mathbf{n}}) v_h \, ds - \alpha \int_{\tilde{\Gamma}} (\nabla v_h \cdot \tilde{\mathbf{n}}) \mathcal{E} u_h \, ds + \int_{\tilde{\Gamma}} \sigma_{\Gamma} \mathcal{E} u_h v_h \, ds = \\ = \int_{\tilde{\Omega}} f v_h \, dx - \int_{\tilde{\Gamma}} (\nabla v_h \cdot \tilde{\mathbf{n}}) g^E \, ds + \int_{\tilde{\Gamma}} \sigma_{\Gamma} g v_h \, ds. \end{aligned} \quad (3)$$

where $\tilde{\mathbf{n}}$ is the outward normal to $\tilde{\Gamma}$ and σ_{Γ} is a penalty parameter.

The choice of the stabilization term, particularly the parameter α , significantly influences the spectral properties of the resulting system matrix. As detailed in our previous work on a Discontinuous Galerkin (DG) based SBM multigrid preconditioner [7], this can be observed even in a simple 1D single-cell problem. For instance, a penalty-free formulation ($\sigma_{\Gamma} = 0$, $\alpha = -1$) can lead to complex eigenvalues, especially when the true boundary lies inside the surrogate domain.

In the context of DG methods, it was found that the choice of stabilization did not have a dramatic impact on the overall multigrid performance, as the cell-based nature of the DG smoother effectively handled local issues [7]. However, in the DG formulation the cellwise smoother experienced significant challenges for $p = 3$. For continuous finite elements, where our smoothers operate on larger, overlapping patches of elements (i.e. vertex or edge patches), the choice of stabilization becomes far more critical. The non-standard terms introduced by SBM are not well-contained within these patches, and an inappropriate stabilization can introduce spectral properties that standard smoothers cannot handle effectively, a problem that is exacerbated for higher-order elements.

The SBM weak formulation with symmetrized stabilization seeks $u_h \in V_h$ such that for all $v_h \in V_h$,

$$\begin{aligned} \int_{\tilde{\Omega}} \nabla u_h \cdot \nabla v_h \, dx - \int_{\tilde{\Gamma}} (\nabla u_h \cdot \tilde{\mathbf{n}}) v_h \, ds - \int_{\tilde{\Gamma}} (\nabla v_h \cdot \tilde{\mathbf{n}}) \mathcal{E} u_h \, ds + \int_{\tilde{\Gamma}} \sigma_{\Gamma} \mathcal{E} u_h \mathcal{E} v_h \, ds = \\ = \int_{\tilde{\Omega}} f v_h \, dx - \int_{\tilde{\Gamma}} (\nabla v_h \cdot \tilde{\mathbf{n}}) g^E \, ds + \int_{\tilde{\Gamma}} \sigma_{\Gamma} g v_h \, ds. \end{aligned} \quad (4)$$

where $\tilde{\mathbf{n}}$ is the outward normal to $\tilde{\Gamma}$ and σ_{Γ} is a penalty parameter. We notice that our discrete solution u_h is a piecewise polynomial function defined on the background mesh $\mathcal{T}_h$, hence its Taylor expansion can be computed directly by evaluating the function values at the points on the true boundary Γ . This allows us to avoid computation of higher-order derivatives of u_h . Note that the resulting form is not symmetric due to the presence of the extrapolated boundary condition.

3 Multigrid Preconditioner

The DG-SBM formulation described above leads to a large, sparse linear system. Due to the lack of symmetry and possible indefiniteness of the resulting matrix, we employ a Krylov subspace method, specifically GMRES, for its solution. The convergence of GMRES is sensitive to the condition number of the system matrix that grows with both the number of elements in the mesh and the polynomial degree. Multigrid methods [4] rely on a hierarchy of meshes with varying resolution levels. Our Cartesian background mesh naturally facilitates the construction of such a nested hierarchy. We define a sequence of meshes $\{\mathcal{T}_{\ell}\}_{\ell=0}^L$, where ℓ denotes the level and L is the finest level. For each level, we identify active cells that have at least a fraction λ of their measure (area in 2D, volume in 3D) inside the domain. On each mesh $\mathcal{T}_{\ell}$, we define a finite element space $\mathbb{V}_{\ell}$ consisting of piecewise polynomials of degree p_{ℓ} . Since the set of active cells is determined by their intersection with the domain boundary, these sets are not necessarily nested between levels, which can affect the efficiency of standard inter-grid transfer operators.

While our construction ensures a nested hierarchy of finite element spaces, the lack of a strict geometric relationship between active cells at different levels may reduce the effectiveness of transfer operators. We mitigate this potential efficiency loss by using only linear elements on geometrically coarser meshes and gradually increasing the polynomial degree only on the finest mesh. Hence the resulting total number of levels is $L + p$.

With this hierarchy of meshes and spaces established, the multigrid method is defined through two crucial components: the smoothers, which reduce high-frequency error components on each level, and the transfer operators, which move information between different resolution levels. These components, together with a coarse-grid solver, form the complete multigrid algorithm. For preconditioning, we use a single multigrid V-cycle.

3.1 Smoother

We first consider Richardson iterations with preconditioners P . Given the current approximation u_h and the right-hand side f_h , a smoothing step updates the approximation:

$$u_h^{k+1} = u_h^k + P(f_h - A_h u_h^k).$$

We decompose the space $\mathbb{V}_\ell$ into a sum of subspaces $\mathbb{V}_i$, i.e., $\mathbb{V}_\ell = \sum_{i=1}^N \mathbb{V}_i$. Let A_i be the restriction of A to the subspace $\mathbb{V}_i$. Then, the additive subspace correction preconditioner is defined as:

$$P = \omega \sum_{i=1}^N A_i^{-1}$$

where ω is a relaxation parameter. Alternatively, the successive subspace correction method is a subspace correction method where the subspaces $\mathbb{V}_i$ are visited in a sequential manner. The procedure can be performed by applying one Richardson iteration with the preconditioner P_i for each subspace $\mathbb{V}_i$. Since in each step only one subspace is corrected, the residual only changes locally and an efficient implementation is possible. The preconditioner updates the solution by sequentially applying corrections for each subspace. For each subspace $\mathbb{V}_i$, a correction is computed and applied:

$$u_h \leftarrow u_h + R_i^T A_i^{-1} R_i (f_h - A_h u_h).$$

This process is repeated for all subspaces $i = 1, \dots, N$. If the subspaces are chosen as one-dimensional spaces spanned by the basis functions, then the preconditioner is equivalent to either Jacobi (additive) or Gauss-Seidel (successive).

Viewing the smoother as a subspace correction method reveals important insights into the convergence properties. The correction computed within a subspace $\mathbb{V}_i$ implicitly satisfies certain boundary conditions on the boundary of that subspace. This is particularly important in the context of the SBM, where the boundary conditions are shifted. A careless choice of subspace can lead to overconstraining the correction, preventing certain error components from being effectively smoothed. For instance, Jacobi and Gauss-Seidel smoothers for continuous finite element methods, which correspond to vertex-patch smoothing with zero correction at patch boundaries, conflict with the shifted boundary conditions and are not effective, as confirmed in our preliminary numerical experiments. In contrast, the cell-wise smoother aligns well with the DG method's element-based structure and takes into account the local nature of the SBM boundary conditions, correcting the solution on each cell independently, as illustrated in Figure 2.

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3.2 Full-Residual Shy Patches

The idea behind our proposed smoother is to construct local problems on patches of cells that are consistent with the global problem, particularly in handling the non-standard terms introduced

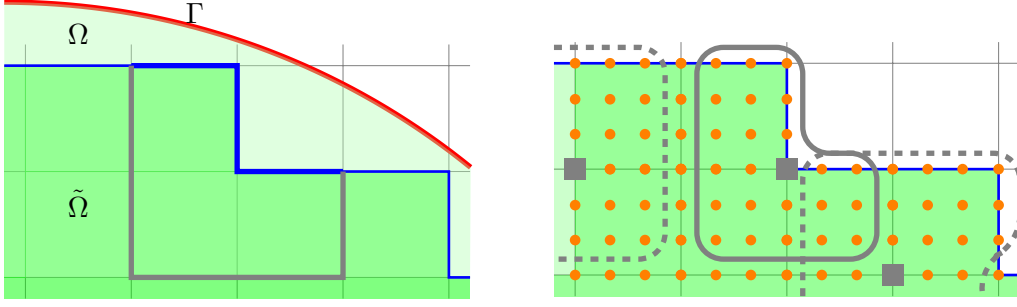


Figure 2: Illustration of a patch. Interior faces ($\mathcal{F}_h^{\text{int}}$, green) connect to neighboring active cells. For cells adjacent to the domain boundary, faces can be part of the surrogate boundary $\tilde{\Gamma}$ (blue), which is shifted from the true boundary Γ (red). The vertex-patch smoother should solve the local problem on patch P ...

by the SBM. We start by constructing vertex patches, where a patch consists of all active cells connected to a given vertex. The subspace for the smoother correction is then defined by the degrees of freedom (DoFs) supported by the cells within this patch.

Next we ensure that the local problem solved on the patch is a faithful representation of the global problem. To achieve this, we only include DoFs for which the complete residual can be evaluated using information solely from the cells within the patch. This means we exclude any DoFs whose support extends to cells outside the current patch. By doing so, we guarantee that the local correction is computed using the true global residual, avoiding inconsistencies that would arise from incomplete information, especially near the surrogate boundary. Figure 2 illustrates this concept for a few patches.

We further refine this patch selection with a concept we call *shyness*. Much like a shy person at a party needs a circle of friends to feel confident enough to join the dance, our vertices are considered shy. A vertex will only form the center of a patch if it is surrounded by a sufficient number of *friends* — in this case, active cells. We define a shyness threshold as the minimum number of active cells required around a vertex to consider it for patch construction. If a vertex does not meet this threshold, it is deemed too isolated, and no patch is formed around it. This strategy prevents the creation of small local problems, which reduces the computational cost of the smoother and avoids potential issues with ill-conditioning that can arise from patches with too few active cells.

In [7], satisfactory results were obtained with 3 smoothing steps. A multi-stage smoothing strategy was proposed in [3] for CutFEM, where an initial global smoothing pass was followed by additional passes restricted to patches containing cut cells. We adopt a similar strategy here. The first smoothing step is applied to all patches in the domain. Subsequent steps are then selectively applied only to those patches that contain cells adjacent to the surrogate boundary. This focuses the computational effort of the smoother on the region where the SBM introduces non-standard terms and where the error is often most difficult to resolve. These boundary-adjacent patches are identified once during the construction of the smoother.

3.3 Transfer Operators

We use the standard L^2 projection operators provided by `deal.II` for the prolongation ($I_{\ell-1}^\ell$) and restriction ($I_\ell^{\ell-1}$) operators, transferring data between coarser and finer grid levels. The restriction operator is the adjoint of the prolongation operator.

While these operators are standard in the context of multigrid methods, we note that the lack of a strict geometric relationship between active cells at different levels may reduce the effectiveness of transfer operators. We tried to reduce this potential efficiency loss by implementing a more sophisticated prolongation that would extrapolate the values that are not defined on the

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coarser mesh, and its adjoint as a restriction. However, this approach did not yield any significant improvements in the convergence rates. We therefore stick to the standard prolongation and restriction operators.

The choice of the threshold parameter λ directly influences the set of active cells, and thus the computational domain $\tilde{\Omega}_\ell$, on each multigrid level ℓ . A lower λ value generally results in more cells being classified as active, potentially leading to a surrogate domain that more closely envelops the true domain. However, the sets of active cells across different multigrid levels, $\tilde{\Omega}_\ell$ and $\tilde{\Omega}_{\ell-1}$, are not guaranteed to be nested. This lack of geometric nestedness can pose a challenge for standard transfer operators. For instance, a coarse cell might be active while some of its children on the finer level are not, or vice-versa. This mismatch can reduce the efficiency of inter-grid transfers, as information might be prolonged to or restricted from regions that are not consistently part of the active domain across levels. While our experiments with more sophisticated extrapolation-based transfer operators did not show significant gains, the interplay between λ , the resulting active cell configurations, and the transfer operator effectiveness remains an important consideration for multigrid performance in SBM.

4 Implementation details

The numerical implementation of our multigrid solver is based on the open-source finite element library `deal.II` [1]. It provides a comprehensive framework for the implementation of finite element methods, including mesh handling, finite element spaces, assembly of linear systems, and interfaces to various linear algebra solvers and preconditioners. Its flexibility and extensive capabilities make it well-suited for developing complex methods like the one presented here, which involves unfitted meshes and multigrid techniques.

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4.1 Background mesh and geometry handling

When implementing SBM on a non-body-fitted mesh, we need to handle cells intersected by the true boundary Γ . We use a level set function $\phi(\mathbf{x})$ to implicitly define the domain $\Omega = \{\mathbf{x} \mid \phi(\mathbf{x}) < 0\}$, with $\Gamma = \{\mathbf{x} \mid \phi(\mathbf{x}) = 0\}$. Cells of the background mesh are classified based on their intersection with the zero level set: cells entirely inside Ω (interior), cells entirely outside Ω (exterior), and cells intersected by Γ . Then, for the intersected cells the fraction of the cell volume inside Ω is computed, and the cell is classified as active if this fraction is greater than a threshold λ . This is handled using non-matching quadrature rules implemented in `deal.II`, which are based on the techniques described in [6].

In our approach, degrees of freedom are formally assigned to all cells of the background mesh, including those that are classified as non-active. However, for non-active cells, the local contribution to the global system matrix is set to the identity matrix, and the corresponding entries in the right-hand side are set to zero. This effectively decouples the degrees of freedom in non-active cells from the system, ensuring that the solution is only computed within the computational domain $\tilde{\Omega}$ while simplifying the global matrix assembly process by maintaining a uniform structure of degrees of freedom (DoF) across all cells.

4.2 Processing surrogate boundary and matrix assembly

The SBM requires computing the closest point projection from points on the surrogate boundary $\tilde{\Gamma}$ to the true boundary Γ . This projection is found by solving a local nonlinear optimization problem for each quadrature point on $\tilde{\Gamma}$, which minimizes the distance to the true boundary defined by the zero level set of a function $\phi(\mathbf{x})$. The full details of the formulation, which involves a Lagrange multiplier approach solved with a Newton-Raphson method, are described in [7]. To ensure sufficient smoothness for the derivative calculations required by the solver, the level set function $\phi(\mathbf{x})$ is represented using finite elements of degree 2 for $p = 1$ and degree p

for $p > 1$. The search for the closest point is performed only in the interior of the cells adjacent to the surrogate boundary. While this does not guarantee that the closest point is found inside the cell, it is expected that even if the closest point is outside the cell, the extrapolation will still yield a good approximation of the boundary condition.

The extrapolation of the function values from the true boundary to the surrogate boundary required for the matrix assembly process is accomplished by evaluating the values of the basis functions at the points on the surrogate boundary. The assembly of the cell contributions and interior faces to the system matrix and right-hand side vector is performed using the standard finite element assembly process provided by `deal.II`.

4.3 Multigrid structures

For the multigrid hierarchy, we leverage `deal.II`'s built-in capabilities for handling nested meshes and defining finite element spaces on each level. The standard L^2 projection operators provided by `deal.II` are used for the prolongation ($I_{\ell-1}^\ell$) and restriction ($I_\ell^{\ell-1}$) operators, transferring data between coarser and finer grid levels.

The assembly of the system matrix A_ℓ on each level ℓ of the multigrid hierarchy follows a procedure analogous to that on the finest level. The level set function defining the domain geometry is interpolated onto the mesh $\mathcal{T}_\ell$. Based on this interpolated level set and the chosen threshold λ , active cells for level ℓ are identified. The SBM bilinear form is then used to assemble the local contributions to A_ℓ only for these active cells. The cells deemed non-active on level ℓ are ignored during the assembly, effectively decoupling their degrees of freedom from the system on that level.

5 Numerical results

In this section, we present numerical results to evaluate the performance of the proposed SBM multigrid preconditioner for the Poisson equation. We investigate its effectiveness in terms of convergence rates, iteration counts under mesh refinement (h -refinement) and polynomial degree increase (p -refinement). To better illustrate the multigrid preconditioner's performance characteristics and enable meaningful comparison of iteration counts, we solve the system to high accuracy with a tolerance of 10^{-12} for the relative residual reduction. This stringent tolerance results in higher iteration counts, making the differences in solver performance more apparent and facilitating clearer assessment of the multigrid effectiveness. Solver failure is defined as exceeding 100 GMRES iterations without reaching this tolerance. The initial background mesh is a square (or cube in 3D) domain covering the range $[-1.01, 1.01]$ in each dimension, subdivided into 4 cells in each coordinate direction. We use $\sigma_\Gamma = 5$ and $\sigma_F = 1$ for the penalty parameters in the SBM formulation. The interior penalty $\sigma_F = 1$ is chosen as a minimal value to ensure coercivity while minimizing its impact on solver convergence, whereas the boundary penalty $\sigma_\Gamma = 5$ is selected to be sufficiently large to ensure stability of the Nitsche coupling on the surrogate boundary.

The majority of our tests are conducted on a unit sphere (a unit disk in 2D, depicted on the left panel of Fig. ??). While geometrically simple, the unit sphere serves as an insightful benchmark. Any sufficiently smooth (C^1) complex boundary, when viewed at a fine enough mesh resolution, locally resembles a flat plane. The unit sphere, due to its uniform curvature, presents a comprehensive range of intersection angles between the true boundary and the background mesh cells. Furthermore, the boundary intersects cells at various locations relative to cell centers and faces, leading to a diverse distribution of shift vector magnitudes and directions. This includes scenarios where the shift vector points from the surrogate boundary towards the interior of the true domain (which we denote as negative shifts if they oppose the outward normal of the surrogate boundary). The right panel of Figure ?? depicts the minimum and

maximum shift magnitudes observed across the surrogate boundary for a typical discretization. The shift magnitudes are normalized by the cell size h ; for a unit cell, the theoretically largest possible shift magnitude in 2D is $\sqrt{2}h$, occurring if the surrogate boundary point is at a cell corner and the true boundary passes through the diagonally opposite corner.

The primary metric for evaluating the multigrid preconditioner is the number of GMRES iterations required to reduce the initial residual by a factor of 10^{-12} . If the threshold is not met within 100 iterations, the solver is considered to have failed. All experiments are conducted using the implementation described in the previous section. As our solver depends on the value of the threshold parameter λ , we will first explore the solver performance on 2D problems as they are less computationally intensive and then show some results for 3D problems with tuned λ .

6 Conclusion

Acknowledgments

The author declares the use of language models (ChatGPT, Gemini, and Claude) to improve the clarity and readability of the manuscript. All scientific content and technical claims are solely the responsibility of the author.

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